COLLIN VAN WEELDEN

319 – 826 – 4366 | collinvw@iastate.edu | Ames, IA (Open to Relocation)

Objective

Seeking to apply hardware development and PCB layout experience in a full-time design engineer position beginning summer 2027

Education

Iowa State University Ames, IA
Bachelor of Science in Electrical Engineering; GPA: 3.90/4.0 *Aug. 2022 – May 2027*

Experience

Quantum Leap Research Leesburg, VA
RF Engineering Intern *May 2026 – Present*

- Built a Python-based analysis and visualization tool for frequency spectrum data, streamlining the team's debugging workflows
- Designed and verified FPGA hardware acceleration to optimize computationally intensive digital signal processing algorithms
- Led the architecture of a custom radio communication protocol, successfully achieving over-the-air data rates up to 150 Kb/s

Iowa State University Ames, IA
Teaching Assistant - Electronic Circuits II (EE 2300) *Jan. 2026 – Present*

- Guided 60+ students through complex analog circuit design concepts and hardware troubleshooting to reinforce core engineering principles
- Evaluated lab assignments and graded exams, providing direct technical feedback to improve student comprehension

Iowa State University Ames, IA
Student Engineer - Electronics and Technology Group *May 2025 – Dec. 2025*

- Engineered a next-generation WiFi board to improve the reliability and scalability of the embedded systems lab
- Diagnosed data integrity issues using oscilloscopes and logic analyzers, and performed component-level PCB repair to restore lab equipment functionality

Collins Aerospace Cedar Rapids, IA
Manufacturing Electrical Engineer - Avionics NPI Team *May 2024 – Dec. 2024*

- Designed multiple custom printed circuit test boards (PCBs) to facilitate aviation product testing, ensuring simplicity and reliability
- Validated initial production units for avionics by executing functional tests and performing hardware troubleshooting and debugging on failures
- Implemented a critical test process solution that expedited the release of 42 past-due units, recovering $2 million in revenue

International Paper Cedar Rapids, IA
Process Controls Engineer *May 2023 – Dec. 2023*

- Interfaced Allen-Bradley PLCs and Honeywell Distributed Controls System (DCS) to ensure safe operation of industrial equipment
- Built and troubleshot over 150 DCS control modules and SCMs using Honeywell Experion PKS

Projects

Drone Electronic Speed Controller (ESC) - Senior Design: Performed schematic design and PCB layout for a 4-layer, high-current custom drone ESC capable of handling 50A burst currents. Designed mixed-signal schematics incorporating STM32 microcontrollers, 3-phase gate drivers, and back-EMF detection for reliable motor control. Dedicated over 150 hours to rigorous circuit design, component selection, and system documentation.

PCB Design Seminar - IEEE: Co-led a hands-on workshop for 27 participants, utilizing industry-standard EDA software for PCB layout and fabrication output generation.

Technical Skills

KiCad, C/C++, Python, MATLAB, Verilog/VHDL, SPICE/Simulations, GNU Radio, SDR, STM32/ESP32, Linux, Git, PCB Soldering, Oscilloscopes, Multimeters